Aidan Hirsch

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EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science in Biomedical Engineering 3.53/4.0 GPA – May 2028

RESEARCH EXPERIENCE

Research Assistant Jan 2026 – Present
Dadarlat Lab – Purdue University *West Lafayette, IN*

- Developing behavior study to encode valence as signed spatial information in ICMS patterns in mouse S1/M1 cortices to investigate human proprioceptive control of bidirectional neuroprostheses
- Modeled and labeled forelimb movements in mouse behavioral videos for DeepLabCut pose estimation model to be used in conjunction with two-photon microscopy neural data for motor cortex analysis
- Rebuilt lab's DeepLabCut analysis pipeline, restored real-time pose estimation for multiple active studies

Research Assistant Jan 2025 – Dec 2025
Neuroprostheses Research Lab – Purdue University *West Lafayette, IN*

- Validated novel correlation between modeled circuit values and electrode degradation by analyzing in-vivo electrical component parameters through UMAP statistical analysis
- Conducted study of ultramicroelectrode array failure modes using 16-week electrochemical impedance spectroscopy recordings in rat somatosensory cortex
- Engineered reusable automation pipeline for data preprocessing, measurement modeling, and file preparation, actively in use and saving 9+ months of manual labor

Research Assistant Apr 2021 – May 2024
Friel Lab – Burke Neurological Institute *White Plains, NY*

- Designed quantitative bimanual hand assessment for children with cerebral palsy, boosting joint location accuracy by 30-50% compared to prior qualitative methods
- Engineered a custom rehabilitation hand-assessment drawer using Raspberry Pi and 3D cameras to capture participants' joint locations in 3D space
- Developed Python program for the drawer assessment that detects participant movement and contact with drawer components by measuring electrostatic discharge

LEADERSHIP AND PROJECTS

Chair Jan 2026 – Present
Purdue IEEE Systems Man and Cybernetics Committee *West Lafayette, IN*

- Leading Purdue's only neurotech club with 55+ active members in development of a 128-channel EEG system, concussion-detection research and publications, and international neural-decoding competitions
- Teaching various aspects of BCI technology, including ICMS, microelectrode arrays, neurotech market developments, neural anatomy, and computational neuroscience on a weekly basis to club members
- Spearheading PCB design workshops for BME and ECE undergraduates to learn how to design PCBs from schematic, to layout, to part selection, to completed board

EEG Build Team Lead Aug 2025 – Present
Purdue IEEE Systems Man and Cybernetics Committee *West Lafayette, IN*

- Directing Electrical team in developing custom 128-electrode EEG system from component level encompassing custom ADC and microcontroller board layouts
- Designed, routed, and fabricated multilayer PCBs integrating 16 ADS1299 ADCs with ESP32-P4 microcontrollers for high-SNR neural signal acquisition
- Implementing an embedded network synchronizing ESP microcontrollers to stream data across all 128 EEG electrode channels

SKILLS

Engineering and Design: KiCad | LTSpice | DeepLabCut | Solidworks | Fusion360 | Blender
Programming and Data Analysis: Python (pandas, NumPy, matplotlib, seaborn) | CEBRA | MATLAB | R | C
Biomedical Techniques: ICMS | Impedance spectroscopy | Rodent handling (IACUC) | Cyclic Voltammetry